Stochastic Calculus and Option Pricing
Lecture 4 Exercise Sheet

4 Week 4 — Theoretical Applications of the Itô Formula

Unless otherwise specified, W is standard Brownian motion, all initial values are deterministic, and coefficients are constant. Work on bounded time horizons and justify each use of a zero-mean stochastic integral.

Part I — Transformations and Their Meaning

Exercise 4.1 (Easy — A transform whose curvature changes sign)**.** Let $Y_t = \log(1 + W_t^2)$.

(a) For $f(x) = \log(1 + x^2)$, compute $f'(x)$ and $f''(x)$. Explain why the classical Itô formula applies on the whole real line.

(b) Find the drift and diffusion coefficient of Y. Determine the regions where its drift is positive, zero, or negative.

(c) Find an explicit function a such that

$$Y_t - \int_0^t a(W_s)\,\mathrm{d}s$$

is a true martingale. Check this by bounding the diffusion coefficient.

(d) Compute $[Y]_t$ as an integral. Explain why the sign changes in the drift do not produce negative increments of quadratic variation.

Exercise 4.2 (Easy to medium — Recovering the noise by an inverse transform)**.** Set $X_t = \sinh(W_t)$, and recall that

$$F(x) = \operatorname{arsinh}(x) = \log\bigl(x + \sqrt{1 + x^2}\bigr).$$

(a) Apply Itô to $\sinh(W_t)$ and express both coefficients entirely in terms of X_t. Use $\cosh^2 w - \sinh^2 w = 1$.

(b) Compute $F'(x)$ and $F''(x)$, then apply Itô to $F(X_t)$. Check explicitly that the drift cancels and the diffusion coefficient is one.

(c) A student keeps only $F'(X_t)\,\mathrm{d}X_t$. Compute the spurious drift produced by that calculation.

(d) Now let $\mathrm{d}Z_t = a(Z_t)\,\mathrm{d}t + \sqrt{1 + Z_t^2}\,\mathrm{d}W_t$. Determine the drift function $a(x)$ for which the drift of $F(Z_t)$ vanishes identically. In that case, identify $F(Z_t)$ from its initial value and differential.

Part II — Which Martingale Claims Survive?

Exercise 4.3 (Easy to medium — The same transform under two dynamics)**.** Let $f \in C^2(\mathbb{R})$ and fix $\mu \neq 0$. Consider, for every starting point $x \in \mathbb{R}$,

$$X_t^x = x + W_t, \qquad Y_t^x = x + \mu t + W_t.$$

(a) Write the drift coefficient of $f(X_t^x)$ and of $f(Y_t^x)$.

(b) Determine all f for which the drift of $f(X_t^x)$ vanishes identically for every x.

(c) Among those functions, determine which also give identically zero drift for $f(Y_t^x)$. Explain why a martingale conclusion depends on the dynamics of the input process.

(d) Verify directly that the candidate processes in parts (b) and (c) are true martingales on bounded horizons, rather than stopping at a formal zero-drift calculation.

Exercise 4.4 (Medium — Correlation between two compensated squares)**.** Let (W, B) be jointly Brownian with $\mathrm{d}[W, B]_t = \rho\,\mathrm{d}t$, $-1 \leq \rho \leq 1$, and set

$$M_t = W_t^2 - t, \qquad N_t = B_t^2 - t.$$

You may use $\mathbb{E}[W_t B_t] = \rho t$ and the fact that Gaussian variables have finite moments of every order.

(a) Find $\mathrm{d}M_t$, $\mathrm{d}N_t$, and $\mathrm{d}[M, N]_t$.

(b) Apply the product rule to $M_t N_t$. Find the finite-variation term that must be subtracted to obtain a true martingale. Justify the integrability of its stochastic terms using Gaussian moments.

(c) Take expectations to compute $\mathbb{E}[M_t N_t]$, then find the correlation of M_t and N_t for $t > 0$. *Hint:* each compensated square has variance $2t^2$.

(d) Explain why changing ρ to $-\rho$ leaves this correlation unchanged. For which ρ is MN itself a martingale? Check the cases $\rho = 0$ and $|\rho| = 1$.

Part III — Products with Accumulated Quantities

Exercise 4.5 (Medium — A random discount factor has finite variation)**.** Define

$$A_t = \int_0^t W_s^2 \, \mathrm{d}s, \qquad D_t = e^{-A_t}, \qquad Z_t = D_t W_t.$$

(a) Find $\mathrm{d}A_t$ and $\mathrm{d}D_t$. Explain why both processes have finite variation on bounded intervals and why $0 < D_t \leq 1$.

(b) Compute $[D, W]_t$ and apply the product rule to Z_t. Explain why randomness of D does not create a mixed correction here.

(c) Find h_t explicitly so that

$$K_t = Z_t + \int_0^t h_s \, \mathrm{d}s$$

is a true martingale starting at zero. Prove the required expected-energy bound for its stochastic integrand.

(d) Compute $[K]_t$ and show that $\mathbb{E}[K_t^2] \leq t$. Does adding back the finite-variation term change quadratic variation?

Exercise 4.6 (Medium — A geometric model with a continuous inflow)**.** Suppose an account receives cash at a constant rate $c > 0$ and satisfies

$$\mathrm{d}X_t = (\mu X_t + c) \, \mathrm{d}t + \sigma X_t \, \mathrm{d}W_t, \qquad X_0 = x > 0.$$

Let $H_t = \exp((\mu - \sigma^2/2)t + \sigma W_t)$, so $\mathrm{d}H_t = \mu H_t \, \mathrm{d}t + \sigma H_t \, \mathrm{d}W_t$ and $H_0 = 1$.

(a) Compute $\mathrm{d}(H_t^{-1})$ and $\mathrm{d}[X, H^{-1}]_t$. Both X and H are driven by the same W.

(b) Apply the product rule to $X_t H_t^{-1}$. Check that its Brownian term and all drift terms proportional to X_t cancel.

(c) Integrate the remaining differential to obtain an explicit expression for X_t in terms of H and an ordinary time integral. Use it to prove $X_t > 0$.

(d) A proposed solution is $\widehat{X}_t = H_t(x + ct)$. Differentiate it and identify the inflow rate it actually implements. Explain why a contribution made at time s must grow by the factor H_t/H_s between s and t.

Part IV — Intensive Itô Formula Practice

Exercise 4.7 (Easy to medium — A time-dependent exponential martingale)**.** For $\lambda \in \mathbb{R}$, define

$$M_t^\lambda = \exp\Big(\lambda W_t - \tfrac{1}{2}\lambda^2 t\Big).$$

(a) Apply the time-dependent Itô formula to $f(t, x) = \exp(\lambda x - \lambda^2 t/2)$, and show that

$$\mathrm{d}M_t^\lambda = \lambda M_t^\lambda \, \mathrm{d}W_t.$$

Identify precisely which two drift contributions cancel.

(b) Prove that M^λ is a true martingale on every bounded horizon. You may compute $\mathbb{E}[(M_t^\lambda)^2]$ from the Gaussian moment-generating function.

(c) Let $a, b \in \mathbb{R}$. Determine all pairs (a, b) for which $\exp(aW_t + bt)$ has zero drift. Explain why the condition is about the quadratic variation of W, not ordinary calculus.

(d) Use conditional expectation to prove $\mathbb{E}[M_t^\lambda \mid \mathcal{F}_s] = M_s^\lambda$ for $s \le t$, directly from independent Brownian increments.

Exercise 4.8 (Medium — A backward equation is a martingale test)**.** Let $u \in C^{1,2}([0,T] \times \mathbb{R})$ have bounded u_x, and define $Z_t = u(t, W_t)$.

(a) Apply Itô's formula and show that Z is a true martingale whenever

$$u_t + \tfrac{1}{2} u_{xx} = 0.$$

Carefully state where boundedness of u_x is used.

(b) Let $g \in C^2(\mathbb{R})$ have bounded first and second derivatives, and set

$$u(t, x) = \mathbb{E}\big[g(x + W_{T-t})\big].$$

Explain, using independent increments and the martingale property from part (a), why $u(t, W_t) = \mathbb{E}[g(W_T) \mid \mathcal{F}_t]$.

(c) For $g(x) = \cos x$, compute u explicitly and verify the backward equation by differentiation.

(d) Explain why the same argument cannot apply directly to $g(x) = |x|$ at $x = 0$, and name the extension of Itô's formula that addresses this kink.